AI Engineer Take-Home Assessment

Company: Halvek Technologies
Product: eSchool
Position: Junior to Mid-Level AI Engineer
Submission Deadline: Monday, 2 November 2026, at 8:00 AM EAT

1. Background

Halvek Technologies is developing **eSchool**, a platform that helps parents discover schools, compare options, and apply for admissions.

Parents can search for schools using different criteria, but not every parent knows exactly which filters to select. We want to make this process easier by allowing parents to describe the kind of school they are looking for in ordinary language.

For this assessment, you will build a small AI-powered school search assistant that translates a parent's request into search criteria and retrieves relevant schools from eSchool.

2. The Task

Build an AI assistant that accepts a parent's natural-language request, identifies the relevant search criteria, and uses eSchool Manager's existing school search functionality to retrieve matching schools.

- **Existing school search page:** https://eschool.ug/find-schools/search
- **Example search URL:**
  `https://eschool.ug/find-schools/search?location=KAMPALA&county=KAMPALA+CENTRAL+DIVISION&min_fees=&max_fees=&gender=mixed`

Example Parent Request:

"I'm looking for a girls' boarding school in Kampala, preferably around Central Division, charging no more than UGX 1 million per term and currently admitting Senior One students."

Your assistant should interpret the request, identify the applicable search criteria, retrieve relevant schools, and present the results clearly.

3. Functional Requirements

A. Interpret the parent's request

The assistant should identify relevant search criteria from natural-language input. Depending on what the parent specifies, these may include:

- Location
- Administrative area or division
- Minimum and maximum fees
- School category or gender
- School level or admission class
- Admission status
- Other filters supported by the existing search functionality

Not every request will contain all these details. Your system should distinguish between specified preferences and unspecified information.

B. Search eSchool

- Use the existing **eSchool** search functionality to retrieve schools based on the interpreted criteria.
- Inspect the available search filters and determine how to integrate with the existing search functionality.
- Do not recreate the school search engine or build an unrelated school dataset.
- You are not expected to have access to the private backend, database, or internal APIs. If the public search interface presents technical limitations, explain those limitations and demonstrate a reasonable alternative.

C. Present the results

- Display the retrieved schools in a clear and understandable format.
- For each school, provide relevant available information and explain why it matches the parent's request.
- Do not invent school details or claim that a school meets a requirement when the available information does not support that conclusion.

D. Handle ambiguity and missing information

Your solution should account for situations such as:

- The parent provides an incomplete or ambiguous request.
- A requested search criterion is unsupported or unavailable.
- Some school information is missing.
- No schools match the requested criteria.
- The search service fails or returns an unexpected response.

Where necessary, the assistant should ask a clarifying question, communicate uncertainty, or explain the limitation rather than silently making assumptions.

E. Demonstrate the solution

Provide examples showing how your assistant handles different requests, including:

1. A request with clearly specified search criteria.
2. A request with missing or ambiguous information.
3. A request that returns no suitable results.

Include the interpreted search criteria and the results produced for each example.

4. Technical Requirements

- Use an AI model or AI API to interpret natural-language requests.
- You may choose your preferred programming language, framework, and AI provider.
- You may implement the solution as a simple web application, API, or command-line application.

- Keep AI interpretation separate from ordinary application logic where appropriate.
- Include basic input validation and error handling.
- Include tests for the main functionality and important edge cases.
- Document any assumptions or limitations encountered when integrating with eSchool Manager.

Note: You do not need to build a production-ready application. The objective is to demonstrate your engineering approach and produce a working prototype within a reasonable scope. Do not include API secrets in your submission.

5. Deliverables

Submit the following:

1. **Source code:** Provide the complete implementation through a GitHub repository or ZIP file.
2. **README:** Include instructions for installing, configuring, and running the application.
3. **Tests:** Include tests covering normal operation and important edge cases.
4. **Design explanation:** Briefly explain:
    - How you designed the solution.
    - How the natural-language request is converted into search criteria.
    - How your application interacts with eSchool.
    - Why you chose your particular AI integration and technical approach.
    - How you handle ambiguity, missing information, and failures.
    - What you would improve if you had more time.
5. **Demonstration:** Provide the example requests and outputs described in Section 3E. Screenshots or a short demonstration video are optional.

6. Assessment Criteria

Criterion	Weight
Problem-solving and correctness	25%
AI integration and appropriate use of AI	20%

Criterion	Weight
System design and technical decisions	20%
Edge-case handling and reliability	15%
Code quality and testing	10%
Documentation and explanation	10%
Total	**100%**

We are interested in the reasoning behind your implementation, not just whether it produces results. A solution that handles ambiguity appropriately, uses the existing search functionality sensibly, and explains its limitations is preferable to one that produces impressive-looking but unreliable recommendations.

7. Submission Instructions

- Deadline: Monday, 2 November 2026, at 8:00 AM East Africa Time (EAT).
- Submission method: Reply to the email through which you received this assessment. Include your GitHub repository link or attach a ZIP file containing your submission.
- Please ensure that your project includes all the information needed to run and evaluate it.
- Late submissions may not be considered unless an extension has been agreed upon in advance.

8. Additional Information

You may consult technical documentation and use AI coding assistants while completing the assignment. However, you should understand the implementation you submit and be prepared to explain your technical decisions, trade-offs, and handling of edge cases during a follow-up discussion.

We do not expect a perfect solution. We want to understand how you approach problems, make engineering decisions, and build practical AI-powered applications.

Good luck!
Halvek Technologies — eSchool Team